

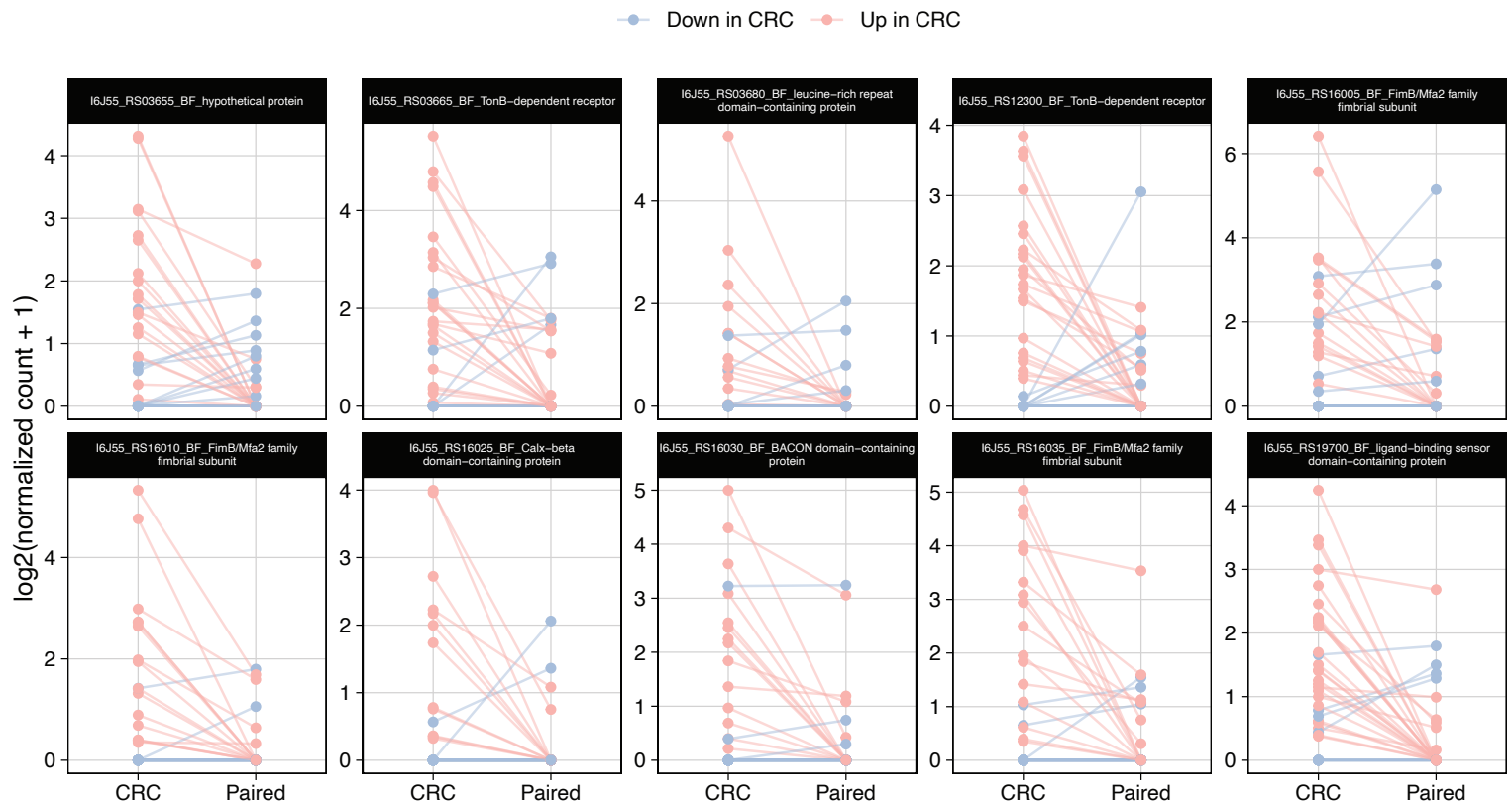


Figure 1. Per-patient expression of the top 10 differentially expressed *B. fragilis* genes between matched tumor and adjacent normal tissue. Mucosal colon biopsies were collected from CRC patients (any T-stage) at Zealand University Hospital. Paired samples consist of a tumor biopsy (CRC) and a biopsy from normal mucosa >10 cm from the tumor (Paired) from the same patient. Host-unmapped RNA-seq reads were processed using SAMSA2, aligned with DIAMOND against the *B. fragilis* reference proteome (NCBI accession GCF_016889925.1). Differential expression was performed with DESeq2, genes were ranked by $\log_2FC$ (CRC vs. Paired), and the top 10 upregulated genes are shown. Each line connects a single patient's tumor-associated CRC and non-tumor associated paired sample. Point colour indicates the direction of change for that patient (red = higher in CRC, blue = lower in CRC). Y-axis shows $\log_2(\text{DESeq2 size-factor normalized count} + 1)$.

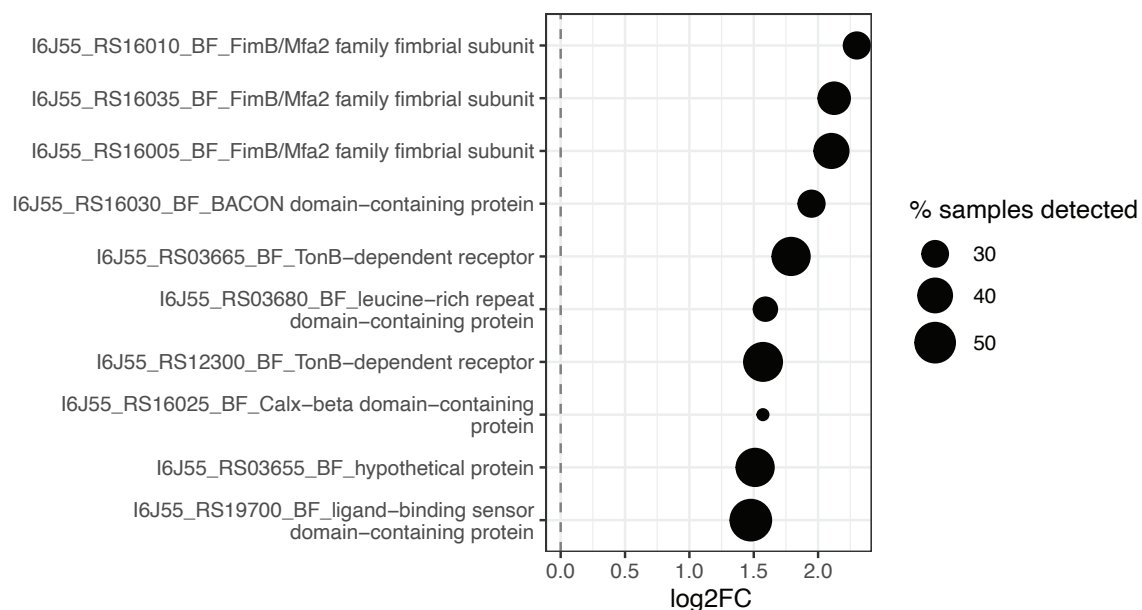


Figure 2. Log2 fold change of the top 10 upregulated *B. fragilis* genes between matched tumor and adjacent normal tissue (CRC vs. Paired) from DESeq2 analysis of host-unmapped RNA-seq reads aligned with DIAMOND against the *B. fragilis* reference proteome (GCF_016889925.1). Genes are ranked by $\log_2FC$, and point size indicates the percentage of samples in which each gene was detected.

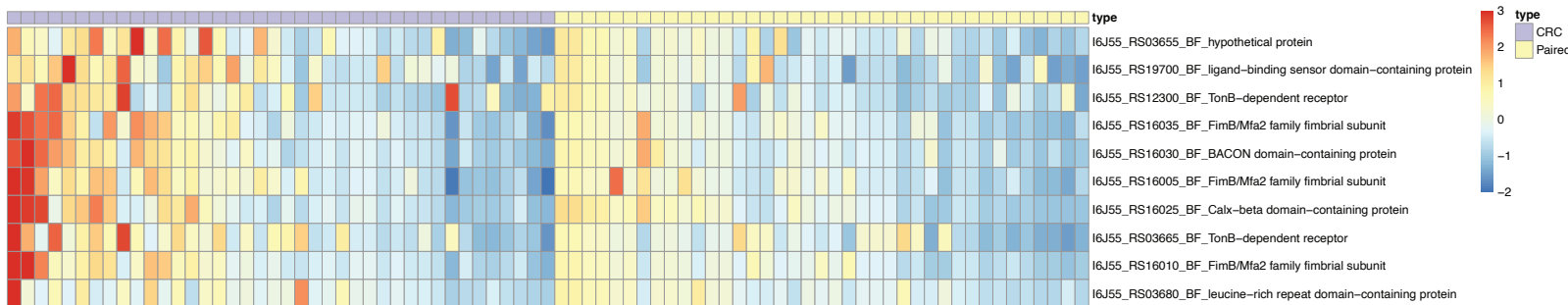


Figure 3. Z-score normalized expression of the top 10 upregulated *B. fragilis* genes across all samples in the CRC vs. Paired comparison. Host-unmapped RNA-seq reads were processed using SAMSA2, then aligned with DIAMOND against the *B. fragilis* reference proteome (NCBI accession GCF_016889925.1). Differential expression was performed with DESeq2; genes were ranked by log2 fold change (CRC vs. Paired) and the top 10 upregulated genes are shown. Columns represent individual samples grouped by tissue type (CRC or Paired) and sorted by descending total z-score within each group. Color reflects row-wise z-score.

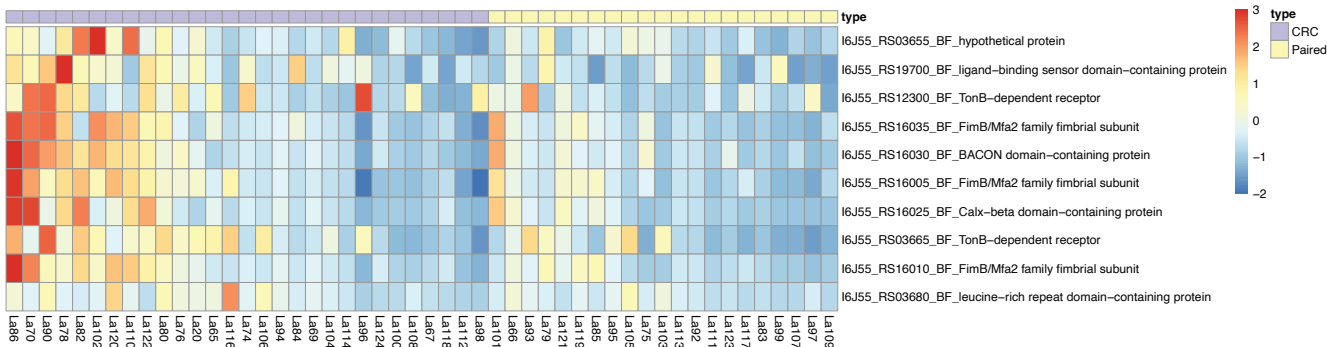


Figure 3 option 2. Filtered to samples with more than 100,000 total unmapped, ribodepleted reads.

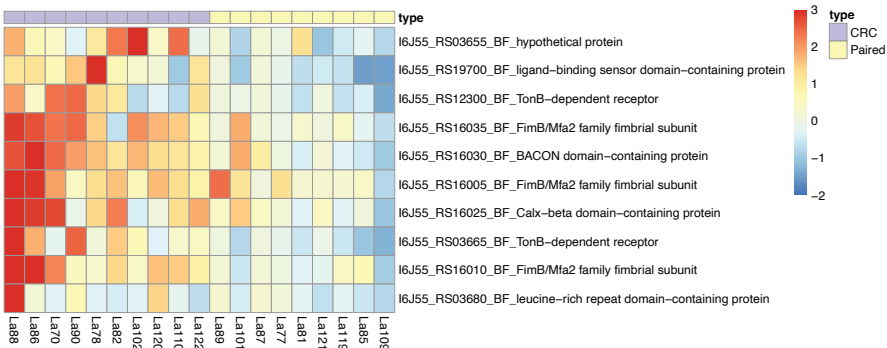


Figure 3 option 3. Filtered to an arbitrary subset of 10 patients.

Comparison 3! This is CRC vs. Healthy (no paired) samples!

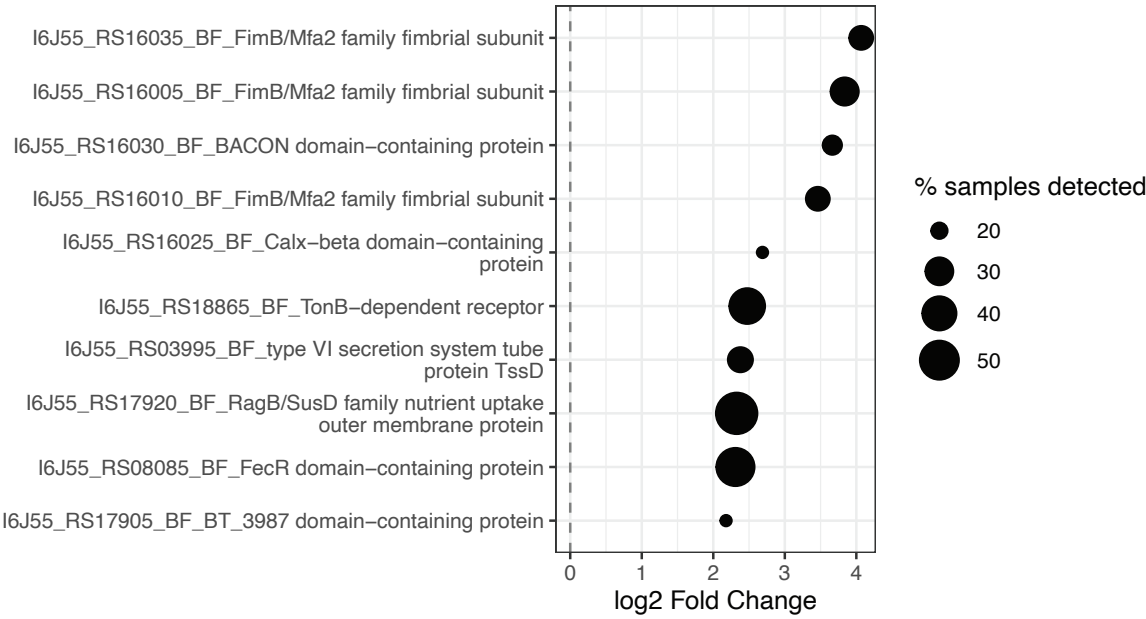


Figure 1. Log2FC of the top 10 upregulated B. fragilis genes between tumor tissue from CRC patients and mucosal colon biopsies from healthy individuals with no observed gastrointestinal diseases (CRC vs. Healthy), from DESeq2 analysis of host-unmapped RNA-seq reads aligned with DIAMOND against the B. fragilis reference proteome (GCF_016889925.1). Genes are ranked by log2 fold change and point size indicates the percentage of samples in which each gene was detected.

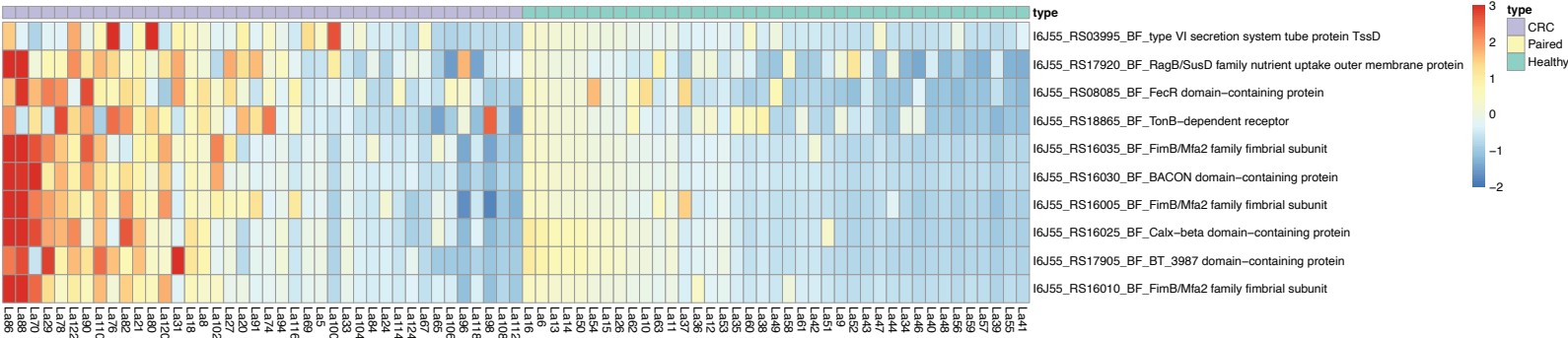


Figure 2. Z-score normalized expression of the top 10 upregulated B. fragilis genes across all samples in the CRC vs. Healthy comparison. Host-unmapped RNA-seq reads were processed using SAMSA2, then aligned with DIAMOND against the B. fragilis reference proteome (NCBI accession GCF_016889925.1). Differential expression was performed with DESeq2; genes were ranked by log2 fold change (CRC vs. Healthy) and the top 10 upregulated genes are shown. Healthy samples are mucosal colon biopsies from individuals with no observed gastrointestinal diseases. Columns represent

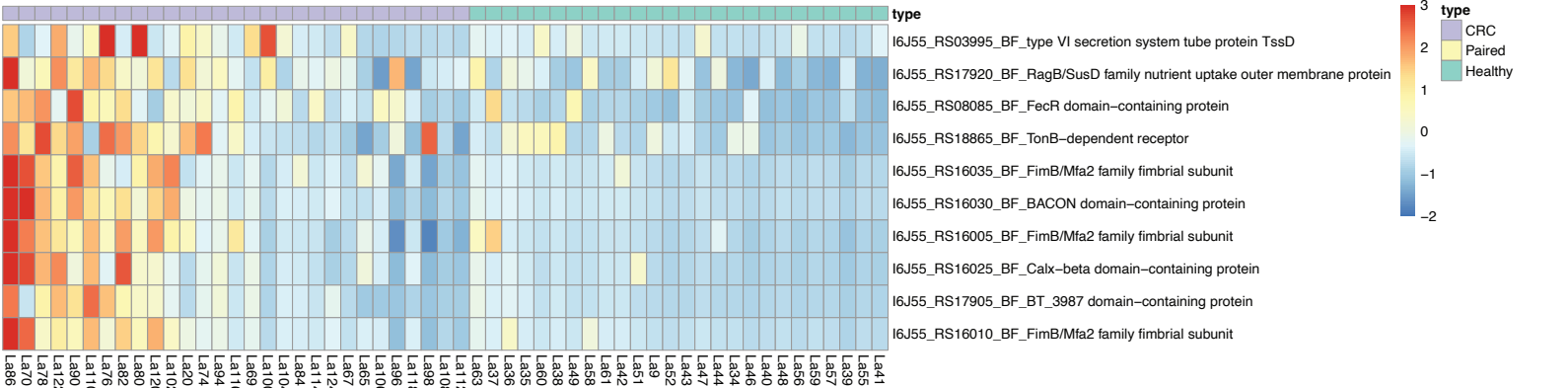


Figure 3 option 2. Filtered to samples with more than 100,000 total unmapped, ribodepleted reads.